

RAL-MoE-ROM: HIERARCHICAL MIXTURE-OF-EXPERTS FOR REGIME-ADAPTIVE REDUCED DYNAMICS

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ABSTRACT

Learning reduced dynamical models over broad parameter ranges is challenging when the underlying system exhibits qualitatively different regimes. A single global representation must simultaneously encode distinct local state geometries, while a single learned dynamical model must reproduce different stability, growth, and saturation mechanisms under recursive rollout. We introduce RAL-MoE-ROM, a two-level hierarchical mixture-of-experts framework for regime-adaptive reduced dynamics. The parameterized solution set is represented by overlapping regime-local reduced charts, each equipped with an autonomous physics–data specialist. Within each specialist, a structured sparse mixture of experts learns corrections to the projected reduced dynamics using nonlinear, affine, and low-rank bilinear expert branches selected by hierarchical sparse routing. A second routing level operates across regime-local specialists. Because the local charts have independent reduced coordinates, adjacent specialists are advanced independently and combined only after reconstruction in the common physical space through a learned Top-2 convex correction. Across three parametrized incompressible-flow benchmarks spanning steady, Hopf-onset, and periodic dynamics, the proposed specialists provide accurate long-horizon autonomous predictions, while the hierarchical fusion consistently improves predictions in evaluated transition regions. Ablations further show complementary contributions from the structured expert parameterization, the projected physical backbone, regime localization, and learned cross-regime routing.

1 INTRODUCTION

Learning reduced-order dynamical models is central to scientific machine learning when high-fidelity simulation is too expensive for repeated evaluation. The problem is especially difficult when a physical parameter changes the qualitative dynamics, not only the response magnitude. Parameterized incompressible flows are representative: different parts of the parameter domain may relax to a steady state, exhibit weak oscillatory growth near instability onset, or sustain nonlinear periodic motion. A many-query reduced model must remain accurate under autonomous rollout across all of these regimes. Unlike a snapshot interpolant, it must continue evolving after the short initial history is exhausted, without injecting reference states into the prediction window.

Broad parameter ranges create two coupled difficulties. First, the state representation is regime dependent: a basis that efficiently describes nearly steady states need not compactly represent developed periodic dynamics. Second, the reduced evolution law must reproduce distinct mechanisms such as decay, growth, phase evolution, and saturation. A global model must solve both problems simultaneously, while recursive prediction continually feeds its own errors back into the dynamics. The difficulty is therefore not only model capacity: global coordinates can spend dimensions reconciling distinct state geometries, and one global evolution law can average mechanisms that should remain regime specific.

Projection-based reduced-order models provide explicit low-dimensional structure through POD and Galerkin projection (Sirovich, 1987; Berkooz et al., 1993; Benner et al., 2015). A projection-based ROM couples a low-dimensional state representation with a reduced dynamical model that advances

its coordinates. Truncated interactions can degrade closure and long-horizon behavior (Amsallem & Farhat, 2012). Learned dynamics offer richer nonlinear maps, yet one-step accuracy need not survive recursive use (Pathak et al., 2018; Vlachas et al., 2018; Duraisamy et al., 2019; Brunton et al., 2020). Hybrid methods therefore combine a projected backbone with learned unresolved terms (Dar et al., 2023; Lee & Carlberg, 2020; Fresca & Manzoni, 2022; Geelen et al., 2023). A complementary approach replaces one global representation with local subspaces, clusters, or manifolds (Kaiser et al., 2014; Li et al., 2021; Diaz et al., 2024). These two remedies address complementary parts of the problem—the first improves the dynamical inductive bias, while the second reduces representation mismatch.

Localization, however, creates an assembly problem. Independently constructed charts have different affine means, bases, scalings, and reduced operators, so their coefficient vectors have no canonical componentwise correspondence. Switching or averaging those vectors can therefore mix incompatible representations. Mixture-of-experts (MoE) models provide a natural language for conditional computation (Jacobs et al., 1991; Jordan & Jacobs, 1994; Shazeer et al., 2017; Fedus et al., 2022), but conventional MoE layers typically route predictors acting in a shared feature space. Regime-dependent reduced dynamics instead require both conditional closure within a chart and selection or combination of complete autonomous models across noncommensurate charts. The outer decision must also respect which local models are supported near a regime boundary; router confidence alone does not make a cross-chart combination dynamically meaningful.

We introduce the *Regime-Adaptive Localized Mixture-of-Experts Reduced-Order Model* (RAL-MoE-ROM), a two-level hierarchy for this setting. Overlapping regime-local charts each support an autonomous physics–data specialist. Within a specialist, hierarchical sparse routing selects structured correction experts with nonlinear, affine, and low-rank bilinear branches while retaining the projected dynamics as an explicit inductive bias. Group Top-1 and channel Top-2 decisions activate a shared expert and a small routed subset for the velocity and pressure channels. Specialists are trained with closed-loop multi-step objectives so that the learned correction is exposed to its own recursive inputs. Across specialists, a parameter-only E2 router produces one trajectory-level regime preference. Physical-history descriptors are used only by the Top-2 convex correction (T2-C) in an admissible overlap. The two selected specialists encode the same physical history in their native charts, evolve independently, and are combined only after reconstruction on the common physical mesh. The fused field is an output and is never fed back into either local rollout.

Our contributions are threefold:

- **Two-level hierarchical routing.** Sparse expert routing models unresolved dynamics inside each local specialist, while a second level selects or combines complete specialists across regimes. This distinguishes conditional closure from conditional selection of an entire dynamical system.
- **Structured closures for autonomous reduced dynamics.** Nonlinear, affine, and low-rank bilinear expert branches augment an explicit projected backbone and are trained under closed-loop rollout. Separate velocity and pressure routing allows the two algebraically different correction channels to adapt independently.
- **Physical-space assembly of noncommensurate local models.** Adjacent specialists retain independent coordinates and rollouts; T2-C couples only their reconstructed fields, avoiding coefficient or operator interpolation. A learned history-conditioned correction refines the parameter-only routing preference while preserving a convex output.

We evaluate RAL-MoE-ROM on three parameterized incompressible-flow systems spanning steady, Hopf-onset, and periodic dynamics. The experiments separate the effects of the structured expert map, projected backbone, localization, and cross-regime assembly. Over the evaluated finite horizons, the local specialists remain accurate and finite, and T2-C improves the corresponding hard or single-specialist prediction in the tested overlaps. The comparisons include vanilla experts, data-only dynamics, a global chart, POD–Galerkin, one-step POD–DeepONet, hard routing, and fixed or probability-based blends. Localization is not uniformly superior to a global model at every parameter; its value emerges from combining local coordinates, regime-specific dynamics, and adaptive assembly.

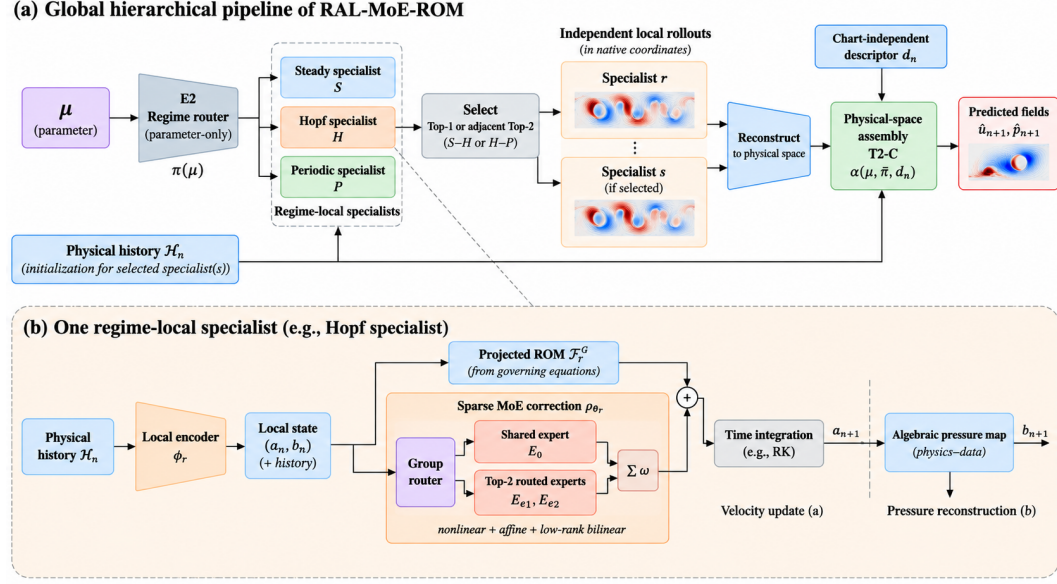


Figure 1: Hierarchical routing in RAL-MoE-ROM. (a) Parameter-only E2 selects one specialist or an admissible adjacent pair. Physical history initializes native charts and supplies T2-C’s descriptor d_n (history-to-assembly line), not E2. Independent rollouts are reconstructed before Top-1 identity output or Top-2 physical-space fusion, without fused-state feedback. (b) Within the selected group, shared and Top-2 experts contribute in parallel. Only velocity is integrated; pressure is reconstructed algebraically from updated velocity and local features (auxiliary connection omitted). Flow thumbnails are schematic, not experimental results.

2 RELATED WORK

Projection, learned dynamics, and autonomous rollout. POD–Galerkin methods project governing equations onto a data-adapted basis and retain an interpretable reduced vector field (Sirovich, 1987; Berkooz et al., 1993; Benner et al., 2015). Their long-horizon accuracy can be limited by truncated scales, imperfect closure, and reduced model stabilization (Amsallem & Farhat, 2012). Non-intrusive models instead learn reduced or latent dynamics (Hesthaven & Ubbiali, 2018; Lee & Carlberg, 2020; Champion et al., 2019; Park et al., 2024), and autoregressive forecasters demonstrate the expressivity of learned evolution laws for complex temporal systems (Pathak et al., 2018; Vlachas et al., 2018). Neural operators learn maps between function spaces (Lu et al., 2021; Kovachki et al., 2023), whereas hybrid ROMs retain physical or projection structure while learning unresolved components (Duraissamy et al., 2019; Brunton et al., 2020; Dar et al., 2023; Fresca & Manzoni, 2022; Geelen et al., 2023). RAL-MoE-ROM follows the hybrid view, but replaces a single residual map with a conditionally sparse correction trained under closed-loop rollout. This distinction is important because low one-step error alone does not establish reliable recursive prediction.

Localized representations and conditional dynamics. Local reduced bases, operator localization, clustering, and nonlinear local manifolds reduce the representational burden imposed on a single global chart (Amsallem & Farhat, 2008; Peherstorfer et al., 2014; Kaiser et al., 2014; Li et al., 2021; Diaz et al., 2024). These approaches motivate parameter-local representations, but localization also makes the dynamics chart dependent: independently constructed models generally differ in their means, bases, scaling, and operators. RAL-MoE-ROM treats each chart as a complete autonomous specialist and does not assume that two local coefficient vectors are commensurate. Its cross-regime problem is therefore not merely basis selection; it is conditional assembly of independently evolved dynamical systems.

Mixture-of-experts and the cross-chart gap. Classical hierarchical mixtures learn conditional expert assignments (Jacobs et al., 1991; Jordan & Jacobs, 1994), and sparse MoE architectures scale conditional computation by activating only a subset of experts (Shazeer et al., 2017; Fedus et al., 2022). Many MoE formulations route predictors or operator blocks within a shared representation or common output space. Recent work extends this specialization to decomposed hard physical constraints (Chalapathi et al., 2024), local implicit representations (Ben-Shabat et al., 2024), shared and routed experts for neural-operator pretraining (MoE-POT; Wang et al. 2025), and operator experts with parameter-conditioned fusion for invariant PDE forecasting (iMOOE; Li et al. 2026). These methods specialize constraints, representations, or operator computations. RAL-MoE-ROM instead places routing at two dynamical levels. Inside a local chart, group and channel routers activate structured closure experts that share the same reduced coordinates. Outside the chart, the E2 router selects complete autonomous projected ROM specialists whose reduced coordinates are independently constructed. An admissible adjacent pair is never averaged in coefficient space: both models roll out independently and T2-C fuses only their reconstructed physical fields. This separation of within-chart conditional closure from cross-chart dynamical assembly is the central gap addressed here.

3 RAL-MoE-ROM

RAL-MoE-ROM combines three operations within a two-level routing hierarchy. Overlapping regime-local charts provide independent reduced representations. Each chart carries a complete autonomous physics–data specialist whose projected dynamics are corrected by a structured sparse MoE. The outer hierarchy selects one specialist or an admissible adjacent pair and assembles only their reconstructed physical fields. Figure 1 summarizes the algorithm; full derivations remain in Appendices A–D.

3.1 OVERLAPPING REGIME-LOCAL REDUCED CHARTS

Let $r \in \{S, H, P\}$ index steady, Hopf-onset, and periodic regimes, and let $\mathcal{M}_r$ denote overlapping parameter regions. A physical velocity field is encoded and reconstructed as

$$\mathbf{a}^{(r)} = \Phi_u^{(r)\top} M_u (\mathbf{u} - \bar{\mathbf{u}}^{(r)}), \quad \hat{\mathbf{u}}^{(r)} = \bar{\mathbf{u}}^{(r)} + \Phi_u^{(r)} \mathbf{a}^{(r)}, \quad (1)$$

where M_u is the finite-volume quadrature matrix, with an analogous map for gauge-fixed pressure coefficients $\mathbf{b}^{(r)}$. A complete chart contains its parameter region, affine means, velocity and pressure bases, local normalization statistics, and projected operators. These quantities are constructed independently for each regime: equal reduced dimensions do not imply corresponding coordinates. This is the reason cross-regime fusion is deferred to physical space.

3.2 PHYSICS–DATA SPECIALISTS WITH STRUCTURED SPARSE EXPERTS

Projection supplies a regime-local dynamical backbone,

$$\dot{\mathbf{a}}^{(r)} = \underbrace{\mathbf{c}_r + \mathbf{A}_r \mathbf{a}^{(r)} + \mathbf{H}_r : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}) + \mathbf{C}_r^p \mathbf{b}^{(r)}}_{\mathcal{F}_r^G(\mathbf{a}^{(r)}, \mathbf{b}^{(r)}; \mu)} + \rho_{\theta_r}^u(\boldsymbol{\xi}^{(r)}). \quad (2)$$

The feature vector $\boldsymbol{\xi}^{(r)}$ contains standardized local coordinates, the projected right-hand side, the parameter, and a short state history. Channel-wise sparse MoEs parameterize the velocity closure $\rho_{\theta_r}^u$ and learned pressure candidate $\rho_{\theta_r}^p$,

$$\begin{aligned} \mathcal{M}_{\theta_r}^c(\boldsymbol{\xi}) &= \omega_0^c E_{0,g^*}^c(\boldsymbol{\xi}) + \sum_{e \in \mathcal{T}_2^c} \omega_e^c E_{e,g^*}^c(\boldsymbol{\xi}), \\ E(\boldsymbol{\xi}) &= N(\boldsymbol{\xi}) + W_{\text{lin}} \boldsymbol{\xi} + \kappa W_o [(U^\top \boldsymbol{\xi}) \odot (V^\top \boldsymbol{\xi})]. \end{aligned} \quad (3)$$

The group router selects a chart-local expert family g^* ; velocity and pressure channels $c \in \{u, p\}$ then use separate Top-2 subsets $\mathcal{T}_2^c$ together with the always-active shared expert. The weights ω_e^c are nonnegative and sum to one over these active experts. In the expert map, N is the nonlinear branch, κ scales the bilinear branch, and $\odot$ denotes componentwise multiplication. The nonlinear,

affine, and low-rank bilinear branches provide complementary corrections without removing the projected backbone. After each velocity macro-step, pressure is reconstructed algebraically as

$$\mathbf{b}_{n+1}^{(r)} = \gamma_n^{(r)} \odot \mathcal{Q}_r^{PP}(\mathbf{a}_{n+1}^{(r)}; \mu) + (\mathbf{1} - \gamma_n^{(r)}) \odot \rho_{\theta_r}^p(\xi_n^{(r)}). \quad (4)$$

Here $\mathcal{Q}_r^{PP}$ is the chart-local pressure–Poisson map, $\rho_{\theta_r}^p$ supplies the learned pressure candidate, and $\gamma_n^{(r)} = \text{sigmoid}(\eta_{\theta_r}^p(\xi_n^{(r)}))$ is a componentwise gate. Velocity is evolved differentially; pressure is recovered by this semi-explicit algebraic closure, not a separate recurrent ODE.

3.3 CLOSED-LOOP AUTONOMOUS TRAINING

Starting from encoded physical history, each specialist recursively consumes its own predictions. For a window of length K , it minimizes the normalized coefficient loss

$$\mathcal{L}_r = \frac{1}{K} \sum_{k=1}^K \left[\frac{\|(\hat{\mathbf{a}}_{n+k}^{(r)} - \mathbf{a}_{n+k}^{(r)}) \odot \sigma_a^{(r)}\|_2^2}{d_u} + \frac{\|(\hat{\mathbf{b}}_{n+k}^{(r)} - \mathbf{b}_{n+k}^{(r)}) \odot \sigma_b^{(r)}\|_2^2}{d_p} \right]. \quad (5)$$

Here d_u, d_p are the local coordinate dimensions and $\sigma_a^{(r)}, \sigma_b^{(r)}$ are training-population scaling vectors; $\odot$ denotes componentwise division. No reference state within the prediction window is injected after initialization. The integrator and horizon curriculum are given in Appendix C; held-out trajectory partitions are in Table 5.

3.4 REGIME ROUTING AND PHYSICAL-SPACE T2-C FUSION

The outer E2 router depends only on the standardized physical parameter $\tilde{\mu}$ and produces a trajectory-level regime preference $\pi(\mu) = \text{softmax}(\mathcal{R}_{E2}(\tilde{\mu}))$. Its probabilities are evaluated once and remain fixed during the rollout. Chart-independent descriptors extracted from the initial physical history $\mathcal{H}_n$ are used only by the T2-C correction gate in an admissible overlap; they are not E2 inputs. A binary mask $m_r(\mathcal{H}_n, \mu, K)$ excludes specialists whose parameter–history pair or rollout horizon is outside the validation-supported region. It constrains the candidate set, not the E2 probability model. Hard Top-1 advances the largest-probability admissible specialist. When an adjacent pair is admissible ($S-H$ or $H-P$), both specialists encode the same physical history in their native coordinates and then evolve independently. For reconstructed fields $\hat{\mathbf{q}}^{(r)}$ and $\hat{\mathbf{q}}^{(s)}$, T2-C predicts a window-conditioned convex weight and forms

$$\hat{\mathbf{q}}_{\text{T2C}}(t) = \alpha \hat{\mathbf{q}}^{(r)}(t) + (1 - \alpha) \hat{\mathbf{q}}^{(s)}(t), \quad \alpha = \sigma(\text{logit}(\bar{\pi}_r) + \Delta_\psi). \quad (6)$$

Here $\mathbf{q}$ collects velocity and gauge-fixed pressure on the common mesh, and σ is the sigmoid. One scalar α is fixed throughout the window and shared by both fields. The pair-normalized E2 probability is $\bar{\pi}_r$, and $\Delta_\psi = c_\psi([\tilde{\mu}; \mathbf{d}_n])$ is a learned correction based on the standardized parameter and chart-independent physical-history descriptor $\mathbf{d}_n$. Fusion occurs after reconstruction and is not fed back into either local rollout. No prediction is asserted if all masks are zero. Admissibility is selected from validation support and should not be interpreted as a formal stability certificate.

4 EXPERIMENTS

We evaluate the two levels of RAL-MoE-ROM through complementary benchmark roles. All three two-dimensional incompressible-flow datasets span steady, Hopf-onset, and periodic dynamics as the Reynolds number varies. The circular-cylinder benchmark provides the local-specialist ablation, the centered-square benchmark provides matched $S-H$ and $H-P$ routing/fusion comparisons, and the fluidic-pinball benchmark provides the autonomous comparison with a classical POD–Galerkin baseline. Table 1 summarizes this scope. All training, validation, and test partitions are defined by complete parameter trajectories; chart-wise ranges and split counts are reported in Appendix E.

Errors E_u and E_p are relative physical-field errors in percent, aggregated according to each benchmark’s evaluation protocol; $E_{\text{joint}} = E_u + E_p$. The sum of displayed component errors can differ slightly from the reported joint error because values are rounded independently. “Divergent” denotes an evaluation window that violates the predefined rollout-divergence criterion, including non-finite states or threshold violations. N/E denotes not evaluable under the prescribed rollout criterion; table-specific notes give the reason.

Table 1: Experimental scope. Local rank pairs (r_u, r_p) and horizons are ordered S, H, P ; K counts rollout steps at each benchmark’s sampling interval. Additional overlap and periodic-core settings are specified in their comparisons.

Benchmark	Re range	Local ranks (S, H, P)	Main horizons	Primary role
Circular cylinder	20–200	$(32, 32), (32, 32), (32, 32)$	56/56/48	Local specialist ablation
Centered square	50–150	$(5, 4), (11, 11), (28, 26)$	16/48/24	Routing and fusion
Fluidic pinball	10–60	$(4, 3), (5, 5), (17, 16)$	56/56/32	Autonomous baseline

Table 2: Autonomous local-specialist prediction on the fluidic-pinball benchmark. Errors are percentages; lower is better. Periodic core ($K = 56$) denotes the longer-horizon stride-four matched final-test evaluation within the periodic regime; complete cadence details are given in Appendix H.

Evaluation setting	Model	E_u	E_p	E_{joint}
Steady ($K = 56$)	POD–Galerkin	N/E [†]	N/E [†]	N/E [†]
	RAL–MoE-ROM	0.2247	0.7593	0.9840
Hopf ($K = 56$)	POD–Galerkin	0.4378 [‡]	48.50 [‡]	48.94 [‡]
	RAL–MoE-ROM	0.0995	1.986	2.085
Periodic ($K = 32$)	POD–Galerkin	1.641	41.56	43.20
	RAL–MoE-ROM	0.2897	0.5518	0.8415
Periodic core ($K = 56$)	POD–Galerkin	2.555	39.36	41.91
	RAL–MoE-ROM	0.6222	1.050	1.672

[†] All 448 evaluation windows violate the predefined rollout-divergence criterion.

[‡] Errors are computed for all numerically finite windows; 141 of 320 Hopf windows exceed the predefined rollout-divergence threshold. All RAL–MoE-ROM settings and both periodic POD–Galerkin settings have zero divergent windows.

4.1 DO LOCAL SPECIALISTS SUPPORT LONG AUTONOMOUS ROLLOUTS?

The learned fluidic-pinball specialists complete all four evaluation settings without divergent windows, whereas POD–Galerkin fails in every steady window and in part of the Hopf evaluation (Table 2). The component errors reveal a particularly large pressure discrepancy for POD–Galerkin; the learned specialists support autonomous prediction over the evaluated horizons. The circular-cylinder ablation and centered-square local-specialist results provide complementary evidence in Table 4 and Appendix G, respectively.

A complementary global POD–DeepONet baseline illustrates the distinction between one-step fitting and autonomous prediction: it achieves $E_u = 1.5921\%$ and $E_p = 23.3447\%$ on 4,228 held-out one-step pairs, yet only one of 28 held-out trajectories remains finite under recursive rollout. Its cadence differs from the matched Periodic specialist evaluation, so this is a one-step versus recursive-stability contrast, not a matched long-horizon accuracy comparison (Appendix H).

4.2 DOES LEARNED PHYSICAL-SPACE FUSION IMPROVE OVERLAP PREDICTION?

To isolate the second routing level, we compare centered-square S – H and H – P overlaps at $K = 24$ (Table 3). Alongside hard routing, equal and E2-probability blends, and the diagnostic oracle, we include T2-C μ -only: the same correction network and optimization protocol with all standardized history-descriptor channels set to zero. This preserves the gate architecture and parameter count while removing history information.

Table 3: Cross-regime routing and fusion on the centered-square benchmark at $K = 24$. Entries are terminal-step E_{joint} (%); lower is better. T2-C μ -only zeroes the standardized history channels in the same gate architecture. The convex oracle is an a posteriori convex reference unavailable during prediction; bold marks the best deployable assembly.

Overlap	Best single*	E2 Top-1	Equal	E2 prob.	T2-C μ -only	T2-C	Oracle
$S-H$	1.7298	1.7298	3.0321	2.1847	2.1846	1.1958	1.1842
$H-P$	1.2079	1.8473	1.6146	1.6618	1.1504	0.8465	0.8384

* Best single is an a posteriori fixed-specialist reference selected by aggregate evaluation error over the corresponding overlap (Hopf for $S-H$, Periodic for $H-P$); it is not a routing strategy available during prediction.

Parameter-only correction behaves differently across overlaps: in $S-H$ it remains close to E2-probability blending (2.1846% versus 2.1847%), whereas in $H-P$ it improves the blend (1.1504% versus 1.6618%). History conditioning provides an additional gain in both, yielding 1.1958% and 0.8465%. Full T2-C reduces joint error by 30.9%/29.9% relative to the best fixed specialist and 30.9%/54.2% relative to E2 Top-1 ($S-H/H-P$), and remains close to the diagnostic convex oracle. Figure 2 shows representative fields and velocity errors.

Across three gate-training seeds, full T2-C gives $E_{\text{joint}} = 1.1957 \pm 0.0001\%$ on $S-H$ and $0.8500 \pm 0.0035\%$ on $H-P$ (sample standard deviation across seeds). Only the gate is retrained; specialists, E2, candidate rollouts, partitions, normalization, and optimization protocol are fixed. Component and seed-wise results are in Appendix G; the fluidic-pinball overlap study is in Appendix H.

4.3 WHAT DO LOCAL ARCHITECTURAL COMPONENTS CONTRIBUTE?

The circular-cylinder comparison isolates three local modeling choices (Table 4). Vanilla-FNN-MoE replaces the structured expert map with conventional feed-forward experts, DataOnly-MoE removes the projected Galerkin and Pressure-Poisson contributions, and Global MoE replaces the local charts and specialists with one global model. The comparison uses physical-field errors so that models with different reduced coordinates can be evaluated on a common quantity. Exact total and per-decision active trainable parameter counts for the analyzed Hopf and Periodic specialists, distinguishing sparse prediction from additional diagnostic evaluations, are reported in Appendix J.

Table 4: Local architecture ablation on the circular-cylinder benchmark. Each regime cell reports the min-max E_u range (upper line) and E_p range (lower line) over held-out Reynolds-number trajectories, in percent. These are field-error ranges, not means or uncertainty intervals. N/E denotes not evaluable under the prescribed rollout criterion: the Vanilla-FNN-MoE Hopf model does not meet the validation-stage stability requirement and is not evaluated on held-out trajectories.

Model	Steady ($K = 56$)	Hopf ($K = 56$)	Periodic ($K = 48$)
RAL-MoE-ROM	0.0330–0.1723	0.0032–0.0288	0.3809–1.3473
	0.2500–1.3249	0.0355–0.4321	1.6825–5.9601
Vanilla-FNN-MoE	0.0384–0.2995	N/E	1.0393–3.0836
	2.2163–18.5603		3.9659–13.3257
DataOnly-MoE	0.1637–0.5122	0.1260–0.3880	0.7854–1.8677
	4751.1025–27275.4645	1280.2811–1524.5949	2.2511–10.8189
Global MoE	0.0845–0.3404	0.0110–0.0268	0.5873–1.1726
	1.5226–7.4363	0.0560–0.2882	2.1896–6.3566

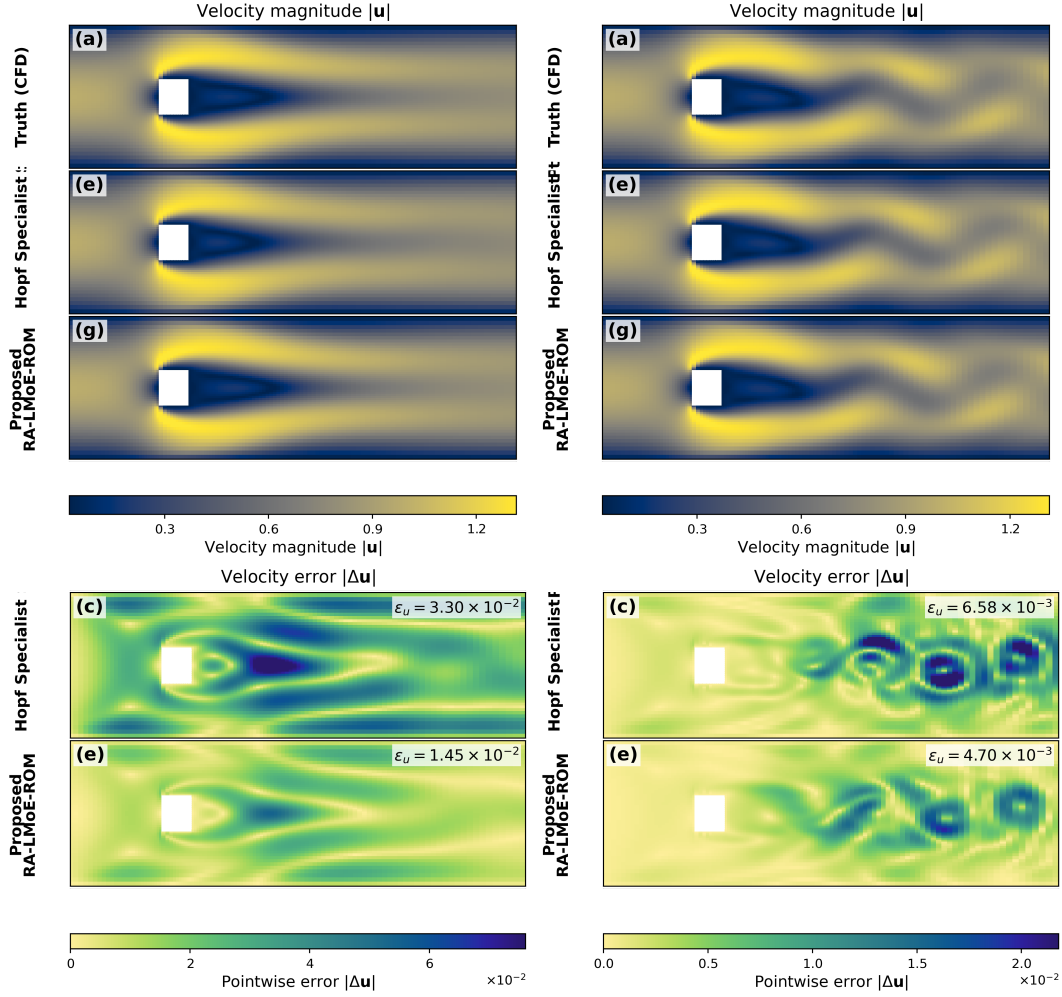


Figure 2: Centered-square $S-H$ (left) and $H-P$ (right) overlap predictions. T2-C fuses independently evolved, reconstructed fields and reduces velocity-field discrepancy relative to the corresponding single specialist. Complete velocity and pressure fields appear in Appendix G.

Structured specialists improve the steady and periodic error ranges over Vanilla-FNN-MoE, which does not meet the validation-stage stability requirement for the Hopf evaluation. Removing the physical backbone particularly degrades steady and Hopf pressure prediction. The global model remains competitive in Hopf and periodic physical-field errors, so localization is not uniformly superior; its role is to provide regime-specific coordinates and dynamics that can participate in cross-regime assembly. The local comparisons address expert structure, physics, and localization, while Table 3 separately evaluates the outer assembly.

Internal routing behavior. Post-hoc analysis of the circular-cylinder Periodic specialist shows conditional use of the Level-1 sparse experts (Figure 3). Across held-out autonomous rollouts, sampled once per macro-step, the velocity channel uses all 18 routed experts through 33 distinct group/pair configurations, while the pressure channel uses 14 of 18 experts through 16 configurations. Group allocation changes across the four held-out Reynolds numbers, and the channels exhibit distinct routing profiles. This diagnostic supports conditional expert utilization, not physical semantics for individual experts; the contrasting Hopf profile is reported in Appendix I.

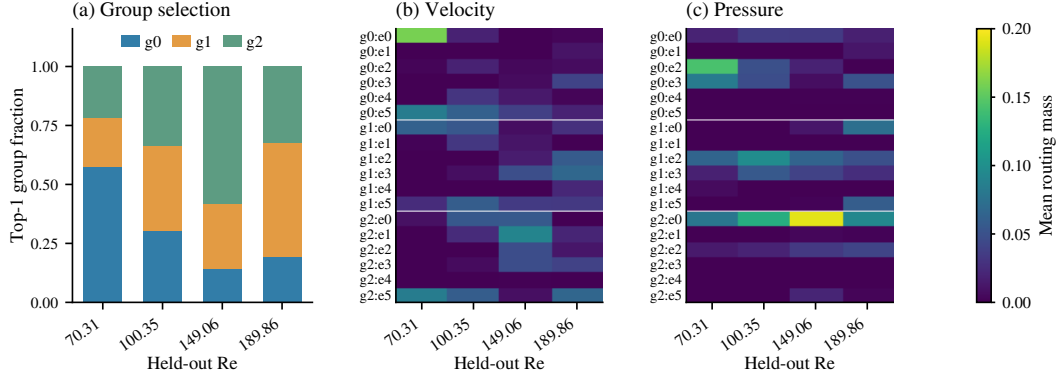


Figure 3: Internal routing of the circular-cylinder Periodic specialist on held-out autonomous rollout windows, with one record per macro-step at the first integration stage. (a) Top-1 group-selection fractions. (b,c) Mean routed-expert mass for velocity and pressure. Expert IDs are local to the specialist, channel, and group; they do not denote physical mechanisms.

5 LIMITATIONS

The regime cover and admissible adjacency graph are selected using validation support; the method does not discover arbitrary bifurcation structure or certify unseen transitions. T2-C is evaluated only between adjacent specialists, and convex physical-space fusion may be insufficient when more than two regimes interact or when conserved quantities require a constrained assembly rule. The evidence concerns two-dimensional incompressible flows and empirical finite-horizon behavior; it does not establish provable stability, conservation, or accuracy outside the sampled parameter ranges. A separate hierarchy is trained for each benchmark, so cross-geometry transfer of one hierarchy is not evaluated. Multiple local bases, operators, and specialists also increase training and model-management cost relative to one global ROM.

Repeated-seed variability is evaluated for the T2-C correction gate with local specialists, E2, and data partitions held fixed. A full multi-seed study of the regime-local specialists and complete hierarchy, and parameter-matched capacity comparisons of the local models, remain future work. The internal-routing analysis is descriptive: it does not establish a unique physical interpretation or a causal performance contribution for individual experts, nor does it imply equally rich expert switching in every chart.

6 CONCLUSION

RAL-MoE-ROM casts broad-parameter reduced dynamics as a two-level mixture-of-experts problem. Structured sparse experts learn unresolved dynamics inside each physics–data specialist, while an outer router selects or combines complete local models. Keeping the local coordinates independent and coupling only reconstructed physical fields avoids interpolation between incompatible modal states. Across the three evaluated flow systems, the specialists achieve low error over the evaluated autonomous rollout horizons, and T2-C improves prediction in the tested transition regions. The results motivate extensions that learn the regime cover, impose physical constraints on field-space fusion, and evaluate the hierarchy on higher-dimensional and non-fluid dynamical systems.

REPRODUCIBILITY STATEMENT

The main text reports the benchmark roles, evaluation horizons, baselines, and metrics. Appendices A–E document the local-coordinate construction, projected operators, specialist architecture, routing and fusion procedure, and implementation details. Appendices F–H provide benchmark-specific results. Appendix I reports post-hoc internal-routing statistics from fixed validation-selected Hopf and Periodic specialists on held-out rollout windows; these diagnostics are not used for model selection. Appendix G additionally reports an architecture-matched descriptor ablation and three-

seed variability of the centered-square T2-C gate, with the underlying specialists and E2 router held fixed.

AI USE STATEMENT

Generative AI tools, including ChatGPT and Codex, assisted with language editing, LaTeX and code refactoring, analysis scripting and interpretation, literature checks, and schematic figure preparation. The authors take full responsibility for reviewing and verifying all scientific content, mathematical formulations, experimental configurations, numerical results, and conclusions, and for the final manuscript and all reported results.

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A FULL PROBLEM SETUP AND WEIGHTED POD CONSTRUCTION

A.1 PARAMETERIZED FULL-ORDER MODEL, GAUGE, AND REGIME COVER

After spatial discretization, the velocity $\mathbf{u}(t; \mu) \in \mathbb{R}^{N_u}$ and pressure $p(t; \mu) \in \mathbb{R}^{N_p}$ satisfy

$$\dot{\mathbf{u}} = \mathbf{f}_h(\mu) + \mathcal{L}_h(\mu)\mathbf{u} - \mathcal{N}_h(\mathbf{u}, \mathbf{u}; \mu) - \nabla_h(\mu)p, \quad \text{div}_h(\mu)\mathbf{u} = 0. \quad (7)$$

Pressure is defined only up to an additive constant. With positive finite-volume weights $\mathbf{w}$, every pressure snapshot is mapped to the weighted zero-mean gauge

$$\Gamma_p(q) = q - \frac{\mathbf{w}^\top q}{\mathbf{w}^\top \mathbf{1}} \mathbf{1}, \quad p^\circ = \Gamma_p(p), \quad \mathbf{w}^\top p^\circ = 0. \quad (8)$$

The parameter domain is covered by overlapping regions

$$\mathcal{M} \subseteq \bigcup_{r \in \{S, H, P\}} \mathcal{M}_r, \quad \mathcal{O}_{SH} = \mathcal{M}_S \cap \mathcal{M}_H, \quad \mathcal{O}_{HP} = \mathcal{M}_H \cap \mathcal{M}_P. \quad (9)$$

No direct S – P overlap is used.

A.2 AREA-WEIGHTED LOCAL POD

The finite-volume quadrature induces

$$\langle \mathbf{v}, \mathbf{z} \rangle_{M_u} = \mathbf{v}^\top M_u \mathbf{z}, \quad \langle q, s \rangle_{M_p} = q^\top M_p s, \quad M_p = \text{diag}(\mathbf{w}), \quad M_u = \text{diag}(M_p, M_p). \quad (10)$$

For chart r , let $X_u^{(r)}$ and $X_p^{(r)}$ contain velocity and gauge-corrected pressure snapshots after subtracting their chart-specific means $\bar{\mathbf{u}}^{(r)}$ and $\bar{p}^{(r), \circ}$. The local bases solve

$$\Phi_u^{(r)} \in \arg \min_{\Psi^\top M_u \Psi = I_{d_u}} \left\| X_u^{(r)} - \Psi \Psi^\top M_u X_u^{(r)} \right\|_{M_u, F}^2, \quad (11)$$

$$\Phi_p^{(r)} \in \arg \min_{\Psi^\top M_p \Psi = I_{d_p}} \left\| X_p^{(r)} - \Psi \Psi^\top M_p X_p^{(r)} \right\|_{M_p, F}^2, \quad (12)$$

where $\|Y\|_{M, F}^2 = \text{tr}(Y^\top M Y)$. Equivalently, for the weighted singular value decompositions

$$M_u^{1/2} X_u^{(r)} = U_u^{(r)} \Sigma_u^{(r)} V_u^{(r)\top}, \quad M_p^{1/2} X_p^{(r)} = U_p^{(r)} \Sigma_p^{(r)} V_p^{(r)\top}, \quad (13)$$

the bases are

$$\Phi_u^{(r)} = M_u^{-1/2} U_u^{(r)}(:, 1:d_u), \quad \Phi_p^{(r)} = M_p^{-1/2} U_p^{(r)}(:, 1:d_p), \quad (14)$$

and satisfy $\Phi_u^{(r)\top} M_u \Phi_u^{(r)} = I$ and $\Phi_p^{(r)\top} M_p \Phi_p^{(r)} = I$. Gauge consistency also gives $\mathbf{w}^\top \Phi_p^{(r)} = \mathbf{0}^\top$.

Encoding and decoding are chart specific:

$$\mathbf{a}^{(r)} = \Phi_u^{(r)\top} M_u (\mathbf{u} - \bar{\mathbf{u}}^{(r)}), \quad \mathcal{D}_r^u(\mathbf{a}^{(r)}) = \bar{\mathbf{u}}^{(r)} + \Phi_u^{(r)} \mathbf{a}^{(r)}, \quad (15)$$

$$\mathbf{b}^{(r)} = \Phi_p^{(r)\top} M_p (\Gamma_p(p) - \bar{p}^{(r),\circ}), \quad \mathcal{D}_r^p(\mathbf{b}^{(r)}) = \bar{p}^{(r),\circ} + \Phi_p^{(r)} \mathbf{b}^{(r)}. \quad (16)$$

Learned components use local standardizations

$$\tilde{\mathbf{a}}^{(r)} = (\mathbf{a}^{(r)} - \boldsymbol{\mu}_a^{(r)}) \oslash \boldsymbol{\sigma}_a^{(r)}, \quad \tilde{\mathbf{b}}^{(r)} = (\mathbf{b}^{(r)} - \boldsymbol{\mu}_b^{(r)}) \oslash \boldsymbol{\sigma}_b^{(r)}. \quad (17)$$

B PROJECTED GALERKIN AND PRESSURE OPERATORS

B.1 CHART-LOCAL MOMENTUM DYNAMICS

Substituting the affine reconstructions into Eq. (7) and testing against the local velocity basis gives

$$\dot{\mathbf{a}}^{(r)} = \mathbf{c}_r(\mu) + A_r(\mu) \mathbf{a}^{(r)} + H_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}) + C_r^p(\mu) \mathbf{b}^{(r)}. \quad (18)$$

For velocity modes $\phi_i^{u,(r)}$ and pressure modes $\phi_\ell^{p,(r)}$,

$$[\mathbf{c}_r]_i = \left\langle \phi_i^{u,(r)}, \mathbf{f}_h + \mathcal{L}_h \bar{\mathbf{u}}^{(r)} - \mathcal{N}_h(\bar{\mathbf{u}}^{(r)}, \bar{\mathbf{u}}^{(r)}; \mu) - \nabla_h \bar{p}^{(r),\circ} \right\rangle_{M_u}, \quad (19)$$

$$[A_r]_{ij} = \left\langle \phi_i^{u,(r)}, \mathcal{L}_h \phi_j^{u,(r)} - \mathcal{N}_h(\bar{\mathbf{u}}^{(r)}, \phi_j^{u,(r)}; \mu) - \mathcal{N}_h(\phi_j^{u,(r)}, \bar{\mathbf{u}}^{(r)}; \mu) \right\rangle_{M_u}, \quad (20)$$

$$[H_r]_{ijk} = - \left\langle \phi_i^{u,(r)}, \mathcal{N}_h(\phi_j^{u,(r)}, \phi_k^{u,(r)}; \mu) \right\rangle_{M_u}, \quad (21)$$

$$[C_r^p]_{i\ell} = - \left\langle \phi_i^{u,(r)}, \nabla_h(\mu) \phi_\ell^{p,(r)} \right\rangle_{M_u}. \quad (22)$$

These terms define the backbone $\mathcal{F}_r^G(\mathbf{a}, \mathbf{b}; \mu)$ used in Eq. (2).

B.2 GAUGE-CONSISTENT PRESSURE-POISSON MAP

At the continuous level the gauge-fixed pressure satisfies

$$-\Delta p^\circ = \nabla \cdot [(\mathbf{u} \cdot \nabla) \mathbf{u} - \mathbf{f}], \quad \int_\Omega p^\circ \, d\Omega = 0, \quad (23)$$

with boundary conditions inherited from the momentum equation. Projection of the discrete relation gives

$$K_r^p(\mu) \mathbf{b}^{PP,(r)} = \mathbf{d}_r(\mu) + B_r(\mu) \mathbf{a}^{(r)} + Q_r(\mu) : (\mathbf{a}^{(r)} \otimes \mathbf{a}^{(r)}), \quad (24)$$

and hence

$$Q_r^{PP}(\mathbf{a}; \mu) = [K_r^p(\mu)]^{-1} [\mathbf{d}_r(\mu) + B_r(\mu) \mathbf{a} + Q_r(\mu) : (\mathbf{a} \otimes \mathbf{a})]. \quad (25)$$

The gauge removes the constant pressure nullspace. Each complete chart therefore contains its own parameter region, affine means, bases, scalars, $\mathcal{F}_r^G$, and Q_r^{PP} . Equal reduced dimensions do not create a canonical componentwise identification between two charts; only decoded physical fields share a common representation.

C SPECIALIST ARCHITECTURE AND CLOSED-LOOP OBJECTIVE

Each regime-local chart is equipped with an independently parameterized semi-explicit specialist. The velocity coefficients evolve under the sum of the chart-local Galerkin vector field and a learned closure,

$$\dot{\mathbf{a}}^{(r)} = \mathcal{F}_r^G(\mathbf{a}^{(r)}, \mathbf{b}^{(r)}; \mu) + \boldsymbol{\rho}_{\theta_r}^u(\boldsymbol{\xi}^{(r)}). \quad (26)$$

Pressure is algebraic rather than an independently evolved recurrent state. After the velocity macro-step, the local pressure–Poisson operator provides the physics-based candidate

$$\mathbf{b}_{n+1}^{PP,(r)} = \mathcal{Q}_r^{PP}(\mathbf{a}_{n+1}^{(r)}; \mu), \quad (27)$$

while a data-driven candidate and a componentwise gate are evaluated from the specialist feature vector,

$$\hat{\mathbf{b}}_{n+1}^{D,(r)} = \rho_{\theta_r}^p(\boldsymbol{\xi}_n^{(r)}), \quad \gamma_n^{(r)} = \text{sigmoid}(\eta_{\theta_r}^p(\boldsymbol{\xi}_n^{(r)})). \quad (28)$$

The next pressure state is the componentwise convex combination

$$\mathbf{b}_{n+1}^{(r)} = \gamma_n^{(r)} \odot \mathbf{b}_{n+1}^{PP,(r)} + (\mathbf{1} - \gamma_n^{(r)}) \odot \hat{\mathbf{b}}_{n+1}^{D,(r)}, \quad \gamma_n^{(r)} \in (0, 1)^{d_p}. \quad (29)$$

C.1 FEATURES AND STRUCTURED SPARSE EXPERTS

Let

$$\mathbf{g}_n^{(r)} = \mathcal{F}_r^G(\mathbf{a}_n^{(r)}, \mathbf{b}_n^{(r)}; \mu) \quad (30)$$

be the current projected right-hand side, and let tildes denote standardization with statistics from the corresponding regime-local training population. The current and two preceding feature blocks are

$$\boldsymbol{\chi}_n^{(r)} = \text{col}(\tilde{\mathbf{a}}_n^{(r)}, \tilde{\mathbf{b}}_n^{(r)}, \tilde{\mathbf{g}}_n^{(r)}, \tilde{\mu}), \quad (31)$$

$$\mathbf{h}_{n,\ell}^{(r)} = \text{col}(\tilde{\mathbf{a}}_{n-\ell}^{(r)}, \tilde{\mathbf{b}}_{n-\ell}^{(r)}, \tilde{\mathbf{g}}_{n-\ell}^{(r)}, \tilde{\mu}), \quad \ell \in \{1, 2\}. \quad (32)$$

Thus the complete input is the three-state history

$$\boldsymbol{\xi}_n^{(r)} = \text{col}(\boldsymbol{\chi}_n^{(r)}, \mathbf{h}_{n,1}^{(r)}, \mathbf{h}_{n,2}^{(r)}). \quad (33)$$

The sampling interval is supplied to the numerical integrator; it is not an additional trainable network feature.

Within each specialist, the group router first makes a hard Top-1 decision,

$$g^\star = \arg \max_{g \in \mathcal{G}} \pi_g^{\text{grp}}(\boldsymbol{\xi}). \quad (34)$$

Velocity and pressure then use separate channel routers. For $c \in \{u, p\}$, let $\mathcal{T}_2^c(\boldsymbol{\xi}; g^\star)$ contain the two routed experts with the largest channel-specific scores. Together with the shared expert $e = 0$, the active output is

$$\mathcal{M}_{\theta_r}^c(\boldsymbol{\xi}) = \omega_0^c E_{0,g^\star}^c(\boldsymbol{\xi}) + \sum_{e \in \mathcal{T}_2^c(\boldsymbol{\xi}; g^\star)} \omega_e^c E_{e,g^\star}^c(\boldsymbol{\xi}), \quad \omega_e^c \geq 0, \quad \omega_0^c + \sum_{e \in \mathcal{T}_2^c} \omega_e^c = 1. \quad (35)$$

Consequently, one group, one shared expert, and two routed experts are active for each output channel. The channel-wise sparse MoEs parameterize the velocity closure $\rho_{\theta_r}^u$ and data-driven pressure candidate $\rho_{\theta_r}^p$. Each expert combines nonlinear, linear/affine, and low-rank bilinear branches,

$$E_{e,g}^c(\boldsymbol{\xi}) = N_{e,g}^c(\boldsymbol{\xi}) + W_{e,g}^{c,\text{lin}} \boldsymbol{\xi} + \kappa W_{e,g}^{c,o} \left[\left(U_{e,g}^{c \top} \boldsymbol{\xi} \right) \odot \left(V_{e,g}^{c \top} \boldsymbol{\xi} \right) \right], \quad q_b = 4, \quad \kappa = 0.05, \quad (36)$$

where q_b is the rank of the bilinear branch and κ scales its contribution to the structured expert map.

C.2 FROZEN-CONTEXT ADVANCEMENT AND CLOSED-LOOP TRAINING

During a velocity macro-step, the current pressure, the recent history, and the physical parameter form a context $\mathcal{C}_n^{(r)}$ that is held fixed across all Runge–Kutta stages. Define

$$F_r(\mathbf{a}; \mathcal{C}_n^{(r)}) = \mathcal{F}_r^G(\mathbf{a}, \mathbf{b}_n^{(r)}; \mu) + \rho_{\theta_r}^u(\boldsymbol{\xi}^{(r)}(\mathbf{a}; \mathcal{C}_n^{(r)})). \quad (37)$$

All three specialists use the classical fourth-order Runge–Kutta update

$$\begin{aligned}
\mathbf{k}_1 &= F_r(\mathbf{a}_n^{(r)}; \mathcal{C}_n^{(r)}), \\
\mathbf{k}_2 &= F_r\left(\mathbf{a}_n^{(r)} + \frac{\Delta t_r(\mu)}{2} \mathbf{k}_1; \mathcal{C}_n^{(r)}\right), \\
\mathbf{k}_3 &= F_r\left(\mathbf{a}_n^{(r)} + \frac{\Delta t_r(\mu)}{2} \mathbf{k}_2; \mathcal{C}_n^{(r)}\right), \\
\mathbf{k}_4 &= F_r\left(\mathbf{a}_n^{(r)} + \Delta t_r(\mu) \mathbf{k}_3; \mathcal{C}_n^{(r)}\right), \\
\mathbf{a}_{n+1}^{(r)} &= \mathbf{a}_n^{(r)} + \frac{\Delta t_r(\mu)}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4).
\end{aligned} \tag{38}$$

The updated velocity is supplied to Eq. (29); the history is then shifted and the autonomous rollout continues.

For a closed-loop rollout of length K , the model recursively consumes its own predicted states and minimizes the regime-locally normalized coefficient error

$$\mathcal{L}_r = \frac{1}{K} \sum_{k=1}^K \left[\frac{1}{d_u} \left\| \left(\hat{\mathbf{a}}_{n+k}^{(r)} - \mathbf{a}_{n+k}^{(r)} \right) \oslash \boldsymbol{\sigma}_a^{(r)} \right\|_2^2 + \frac{1}{d_p} \left\| \left(\hat{\mathbf{b}}_{n+k}^{(r)} - \mathbf{b}_{n+k}^{(r)} \right) \oslash \boldsymbol{\sigma}_b^{(r)} \right\|_2^2 \right]. \tag{39}$$

Training minimizes Eq. (39) over progressively longer rollout horizons $K \in \{4, 8, 12, 16\}$. The three specialists are trained and selected independently; subsequent outer-router or T2-C training does not update their local bases, scalars, reduced operators, internal routers, experts, integrators, or pressure maps.

D OUTER ROUTING AND T2-C DETAILS

Parameter-only routing and admissibility. The E2 router receives only the standardized scalar parameter,

$$\tilde{\mu} = \frac{\mu - \mu_R}{\sigma_R}, \quad \boldsymbol{\pi}(\mu) = \text{softmax}(\mathcal{R}_{E2}(\tilde{\mu})) = [\pi_S, \pi_H, \pi_P]^\top. \tag{40}$$

The trajectory-level probabilities are evaluated once and remain unchanged during rollout. Adjacency is fixed by

$$A_{\text{adj}} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad \mathcal{C}_{\text{top}}(k) = \{j : A_{\text{adj},kj} = 1\}, \tag{41}$$

with ordering (S, H, P) , so direct S – P fusion is excluded. If $m_j(\mathcal{H}_n, \mu, K) \in \{0, 1\}$ denotes validation-supported dynamical admissibility, the masked preference is

$$\tilde{\pi}_j = \frac{\mathbf{1}_{\{j \in \mathcal{C}_{\text{top}}\}} m_j \pi_j}{\sum_{k \in \{S, H, P\}} \mathbf{1}_{\{k \in \mathcal{C}_{\text{top}}\}} m_k \pi_k + \varepsilon}. \tag{42}$$

One admissible candidate gives a Top-1 prediction; an admissible adjacent pair is passed to T2-C. No ROM prediction is asserted when every mask is zero. For a pair $(1, 2)$, define

$$\bar{\pi}_1 = \frac{\pi_1}{\pi_1 + \pi_2}, \quad \bar{\pi}_2 = 1 - \bar{\pi}_1, \quad \ell^{E2} = \log \frac{\bar{\pi}_1 + \varepsilon}{\bar{\pi}_2 + \varepsilon}. \tag{43}$$

Chart-independent physical-history descriptor. T2-C, unlike E2, uses the current physical state and two preceding states. Let $|\Omega| = \mathbf{1}^\top \mathbf{w}$, $\Delta t_n = t_n - t_{n-1}$, and let U_{ref} , P_{ref} , and T_{ref} be fixed reference scales. With gauge-corrected pressure p_n° , define

$$\hat{E}_n^u = \frac{\|\mathbf{u}_n\|_{M_u}^2}{|\Omega| U_{\text{ref}}^2}, \quad \hat{E}_n^p = \frac{\|p_n^\circ\|_{M_p}^2}{|\Omega| P_{\text{ref}}^2}, \tag{44}$$

$$\hat{D}_n^u = \frac{T_{\text{ref}} \|\mathbf{u}_n - \mathbf{u}_{n-1}\|_{M_u}}{\Delta t_n \sqrt{|\Omega|} U_{\text{ref}}}, \quad \hat{D}_n^p = \frac{T_{\text{ref}} \|p_n^\circ - p_{n-1}^\circ\|_{M_p}}{\Delta t_n \sqrt{|\Omega|} P_{\text{ref}}}, \tag{45}$$

$$\hat{g}_n = \frac{T_{\text{ref}}}{\Delta t_n} \log \frac{\hat{E}_n^u + \varepsilon_E}{\hat{E}_{n-1}^u + \varepsilon_E}. \tag{46}$$

The second-order velocity indicator is

$$\mathbf{A}_n^u = \frac{2}{\Delta t_n + \Delta t_{n-1}} \left(\frac{\mathbf{u}_n - \mathbf{u}_{n-1}}{\Delta t_n} - \frac{\mathbf{u}_{n-1} - \mathbf{u}_{n-2}}{\Delta t_{n-1}} \right), \quad (47)$$

$$\hat{\kappa}_n^u = \frac{T_{\text{ref}}^2 \|\mathbf{A}_n^u\|_{M_u}}{\sqrt{|\Omega|} U_{\text{ref}}}. \quad (48)$$

The six statistics $\mathbf{s}_n = [\hat{E}_n^u, \hat{E}_n^p, \hat{D}_n^u, \hat{D}_n^p, \hat{g}_n, \hat{\kappa}_n^u]^\top$ are standardized as

$$\mathbf{d}_n = (\mathbf{s}_n - \boldsymbol{\mu}_s) \oslash (\boldsymbol{\sigma}_s + \varepsilon_s \mathbf{1}) \in \mathbb{R}^6. \quad (49)$$

Because these quantities are computed from physical fields, they do not depend on either candidate’s POD coordinates.

Correction gate and objective. The gate input $[\tilde{\mu}; \mathbf{d}_n] \in \mathbb{R}^7$ is processed by two hidden layers of width 64 with SiLU activations and a scalar linear output. The output layer is zero-initialized, so training begins from the E2 pair probability. The gate predicts

$$\Delta_\psi = c_\psi([\tilde{\mu}; \mathbf{d}_n]), \quad \alpha = \text{sigmoid}(\ell^{E2} + \Delta_\psi). \quad (50)$$

Both specialists start from the same three-state physical history, encode it in their native charts, and perform independent autonomous rollouts. The scalar α is fixed over the prediction window. Velocity and gauge-corrected pressure are reconstructed on the common mesh and combined convexly; the fused output is never projected back into either rollout.

For candidate field errors $\mathbf{e}_1^c$ and $\mathbf{e}_2^c$, $c \in \{u, p\}$, define

$$Q^c(\alpha) = \alpha^2 \|\mathbf{e}_1^c\|_{M_c}^2 + (1 - \alpha)^2 \|\mathbf{e}_2^c\|_{M_c}^2 + 2\alpha(1 - \alpha) \langle \mathbf{e}_1^c, \mathbf{e}_2^c \rangle_{M_c}. \quad (51)$$

The correction map minimizes

$$\mathcal{L}_{\text{T2C}} = \text{mean}_{w,k} \left[\frac{Q^u(\alpha^{(w)})}{\|\mathbf{u}_{\text{true}}\|_{M_u}^2 + \varepsilon} + \frac{Q^p(\alpha^{(w)})}{\|p_{\text{true}}\|_{M_p}^2 + \varepsilon} \right]. \quad (52)$$

Gate checkpoints are selected on validation data by lexicographically minimizing the worst Reynolds-number mean error and then the global mean error. The equal blend fixes $\alpha = 1/2$; the probability blend sets $\alpha = \bar{\pi}_1$; and the convex oracle chooses its weight a posteriori for each evaluation window. The oracle is a reference, not a deployable method.

Centered-square convex oracle. For each evaluation window, the diagnostic oracle searches $\alpha \in \{0, 10^{-4}, \dots, 1\}$ and selects the value minimizing mean $E_u + E_p$ over all 24 rollout steps. Table 3 then reports the terminal-step joint error with that window-wise weight. Thus the oracle uses the reference solution a posteriori but is not defined as the exact minimizer of the reported terminal-step metric; its objective also differs from the squared-error gate-training loss.

E REPRODUCIBILITY AND IMPLEMENTATION DETAILS

Trajectory-level data partition and evaluation protocol. The split unit is a complete Reynolds-number trajectory, not an individual snapshot or a randomly sampled time window. Each local specialist is fitted only on its stated training trajectories, model and routing choices use the validation trajectories, and test trajectories are used only for the reported autonomous rollouts. A Reynolds number that occurs in two overlapping fluidic-pinball charts keeps the same split role in both.

Unified autonomous integrator. All steady, Hopf, and periodic specialists use the same classical fourth-order Runge–Kutta update with a frozen context within each macro-step. Specifically, the pressure coefficients, recent-history features, and physical parameter are held fixed across the four RK stages; only the trial velocity state changes from stage to stage. After the RK4 velocity update, the algebraic pressure closure is evaluated, the history is shifted, and the resulting predictions are fed back at the next macro-step. The rollout is initialized from the available physical history, and no reference state from inside the prediction window is injected thereafter.

Table 5: Chart-wise parameter coverage, trajectory counts, and rollout horizons. Train, validation, and test counts refer to complete Reynolds-number trajectories.

Benchmark	Chart	Re range	Train	Validation	Test	K
Circular cylinder	S	20.0–46.4	14	2	4	56
	H	45.5–59.2	29	2	3	56
	P	57.3–200	53	6	4	48
Centered square	S	50–95.4	60	5	4	16
	H	94–102	23	6	5	48
	P	98.5–150	27	6	4	24
Fluidic pinball	S	10–19	37	8	7	56
	H	16.6–25	39	8	7	56
	P	21.25–60	47	9	9	32

Datasets and local reduced dimensions. For the circular cylinder, all three charts use $(r_u, r_p) = (32, 32)$ and their regime-specific snapshot intervals. For the centered square, (r_u, r_p) is $(5, 4)$, $(11, 11)$, and $(28, 26)$ for the steady, Hopf, and periodic charts, respectively. Its 120 complete trajectories cover $Re \in [50, 150]$ on a 9,400-cell finite-volume mesh; the CFD step is $\delta t_{\text{CFD}} = 0.02$, and one snapshot is recorded every 200 CFD steps, so $\Delta t_{\text{snap}} = 4$. For fluidic pinball, the corresponding ranks are $(4, 3)$, $(5, 5)$, and $(17, 16)$. Its 131 complete trajectories cover $Re \in [10, 60]$, with 94/19/18 train/validation/test trajectories overall, on a 70,194-cell mesh; $\delta t_{\text{CFD}} = 0.005$ and snapshots are sampled every 0.25 physical time units. Chart-wise Reynolds-number ranges, split counts, and rollout horizons are listed in Table 5.

Admissible routing and fusion. The outer regime graph permits only adjacent S – H and H – P pairs; neither a direct S – P pair nor a three-specialist route is evaluated. The parameter-only E2 probabilities are evaluated once for a complete fixed- Re trajectory and remain fixed throughout its rollout. In an admissible Top-2 case, the two specialists encode the same initial physical history, evolve independently in their native coordinates, and are combined only after reconstruction in the common physical space. Rollout windows and routing/fusion choices are fixed using validation data before test evaluation.

Reproducibility limitations. The main comparisons retain fixed validation-selected runs. Appendix G adds a parameter-matched gate descriptor ablation and three-seed gate variability with specialists, E2, partitions, and normalization fixed. Multi-seed evaluation and parameter-matched capacity controls for the local specialists remain future work.

F CIRCULAR-CYLINDER RESULTS

Table 6 reports the min–max range over held-out long-horizon trajectories. “N/E” denotes not evaluable under the prescribed rollout criterion.

In the S – H overlap, E2 Top-1 gives $E_u = 0.0357$ – 0.0410% and $E_p = 1.6746$ – 1.9028% . T2-C reduces these ranges to 0.0148 – 0.0355% and 0.7387 – 1.7212% , with mean steady weight spanning 0.0861 – 0.9987 . The global model yields $E_u = 0.0851$ – 0.0953% and $E_p = 1.9517$ – 2.3106% on the same overlap evaluation.

Table 6: Circular-cylinder long-horizon error ranges (%). The Vanilla-FNN-MoE Hopf model does not meet the validation-stage stability requirement and is therefore not evaluated on held-out trajectories (N/E).

Regime	Model	E_u	E_p
Steady	RAL-MoE-ROM	0.0330–0.1723	0.2500–1.3249
	Vanilla-FNN-MoE	0.0384–0.2995	2.2163–18.5603
	DataOnly-MoE	0.1637–0.5122	4751.1025–27275.4645
	Global MoE	0.0845–0.3404	1.5226–7.4363
Hopf	RAL-MoE-ROM	0.0032–0.0288	0.0355–0.4321
	Vanilla-FNN-MoE	N/E	N/E
	DataOnly-MoE	0.1260–0.3880	1280.2811–1524.5949
	Global MoE	0.0110–0.0268	0.0560–0.2882
Periodic	RAL-MoE-ROM	0.3809–1.3473	1.6825–5.9601
	Vanilla-FNN-MoE	1.0393–3.0836	3.9659–13.3257
	DataOnly-MoE	0.7854–1.8677	2.2511–10.8189
	Global MoE	0.5873–1.1726	2.1896–6.3566

G CENTERED-SQUARE RESULTS

Table 7 separates the velocity and pressure errors for the joint-field comparison in Table 3. Best single in the main text means one fixed specialist per overlap, chosen for comparison by the lowest aggregate joint error: Hopf for $S-H$ and Periodic for $H-P$. It is distinct from the window-wise convex oracle, which uses the reference solution to choose a weight for each evaluation window.

Overlap evaluation and descriptor ablation. The routing/fusion errors E_u , E_p , and $E_{\text{joint}} = E_u + E_p$ are terminal-step ($k = 24$) errors averaged over 16 held-out windows per overlap. Each of the two held-out Reynolds numbers contributes eight windows, so the pooled window mean equals the mean of the per- Re window averages. This terminal metric is distinct from the full-window mean used by the oracle (Appendix D). T2-C μ -only retains the correction network, parameter count, training/validation partitions, optimizer, and selection rule, but sets all standardized history-descriptor channels to zero. Table 7 retains the original fixed-run comparisons and adds the μ -only run with seed 42001; seed averages are reported separately.

Table 7: Complete centered-square routing and fusion errors at $K = 24$. All errors are percentages. The convex oracle is an a posteriori reference; bold marks the best deployable assembly for each field metric. T2-C μ -only zeroes standardized history channels without changing the gate architecture.

Overlap	Predictor	E_u	E_p	E_{joint}
$S-H$	Steady specialist	2.5976	2.2450	4.8426
	Hopf specialist	0.4942	1.2356	1.7298
	E2 Top-1	0.4942	1.2356	1.7298
	Equal blend	1.3644	1.6677	3.0321
	E2-probability blend	0.7898	1.3949	2.1847
	T2-C μ -only	0.7898	1.3948	2.1846
	T2-C	0.4615	0.7343	1.1958
	Convex oracle	0.4659	0.7183	1.1842
$H-P$	Periodic specialist	0.6420	0.5659	1.2079
	Hopf specialist	1.0034	1.8684	2.8718
	E2 Top-1	0.7422	1.1052	1.8473
	Equal blend	0.6077	1.0069	1.6146
	E2-probability blend	0.6154	1.0464	1.6618
	T2-C μ -only	0.5875	0.5629	1.1504
	T2-C	0.4509	0.3956	0.8465
	Convex oracle	0.4479	0.3905	0.8384

The aggregate local-specialist errors are $(E_u, E_p) = (1.5311, 1.4583)\%$ for steady flow, $(0.4387, 0.4844)\%$ for Hopf-onset flow, and $(0.9395, 1.0921)\%$ for periodic flow. On the aggre-

Table 8: Parameter-wise centered-square routing and T2-C results at $K = 24$. Joint errors and relative error reductions are percentages. The mean weight is the steady weight for $S-H$ and the periodic weight for $H-P$.

Overlap	Re	E2 choice	$\bar{\alpha}$	E2 E_{joint}	T2-C E_{joint}	Reduction
$S-H$	95.1	Hopf	0.0967	1.7140	1.1827	31.00%
$S-H$	95.3	Hopf	0.0971	1.7456	1.2090	30.74%
$H-P$	100.5	Hopf	0.5870	2.6485	0.8571	67.64%
$H-P$	102.0	Periodic	0.7094	1.0462	0.8360	20.09%

gate $S-H$ evaluation, the steady and Hopf specialists have joint errors 4.8426% and 1.7298%; on $H-P$, the periodic and Hopf specialists give 1.2079% and 2.8718%. These results show why the best local choice can change between overlaps and why a corrected convex assembly is useful.

Gate-training seed variability. Only the lightweight gate is retrained for seeds 42001, 42002, and 42003. Specialists, E2, candidate rollouts, partitions, normalization, and optimization settings are fixed. Both modes use AdamW (learning rate 3×10^{-4} , weight decay 10^{-4}), 8,000 steps, and gradient-norm clipping at 1. Minibatches sample training windows with replacement. Validation is evaluated at step 1 and every 100 steps; the selected model minimizes the worst- Re full-window mean joint error, then the pooled mean, without test-based selection. Table 9 reports mean and sample standard deviation ($N = 3$, denominator $N - 1$); this is gate-seed variability, not across- Re variation, a confidence interval, or full-pipeline uncertainty. Table 10 gives individual joint errors.

Table 9: Three-seed variability of the centered-square T2-C gate at the K24 terminal step. Values are percentages, mean $\pm$ sample SD across gate-training seeds, conditional on fixed specialists, E2, candidate rollouts, partitions, and normalization. Bold marks the lower mean within each overlap.

Overlap	Gate	E_u	E_p	E_{joint}
$S-H$	T2-C μ -only	0.79006 $\pm$ 0.00026	1.39498 $\pm$ 0.00012	2.18504 $\pm$ 0.00038
	Full T2-C	0.46156 $\pm$ 0.00004	0.73416 $\pm$ 0.00014	1.19572 $\pm$ 0.00010
$H-P$	T2-C μ -only	0.59067 $\pm$ 0.00461	0.56050 $\pm$ 0.00435	1.15116 $\pm$ 0.00070
	Full T2-C	0.45211 $\pm$ 0.00138	0.39793 $\pm$ 0.00215	0.85004 $\pm$ 0.00347

Table 10: Seed-wise centered-square K24 terminal E_{joint} (%). Each column uses the same training seed for both gate variants.

Overlap	Gate	42001	42002	42003
$S-H$	T2-C μ -only	2.18460	2.18524	2.18527
	Full T2-C	1.19584	1.19567	1.19565
$H-P$	T2-C μ -only	1.15037	1.15170	1.15143
	Full T2-C	0.84653	0.85011	0.85348

Full T2-C improves upon μ -only for all three paired seeds in both overlaps; the reductions in three-seed mean joint error are 45.28% ($S-H$) and 26.16% ($H-P$). The $S-H$ μ -only runs all select step 1 after completing 8,000 training steps; later models do not improve the prescribed validation criterion. They remain close to direct probability blending. The $H-P$ μ -only runs select step 400 and improve that baseline. These observations support history conditioning under the tested protocol, not a universal limitation of parameter-only correction.

Figures 4 and 5 provide the complete velocity, gauge-corrected pressure, and pointwise-error views underlying the compact main-text comparison.

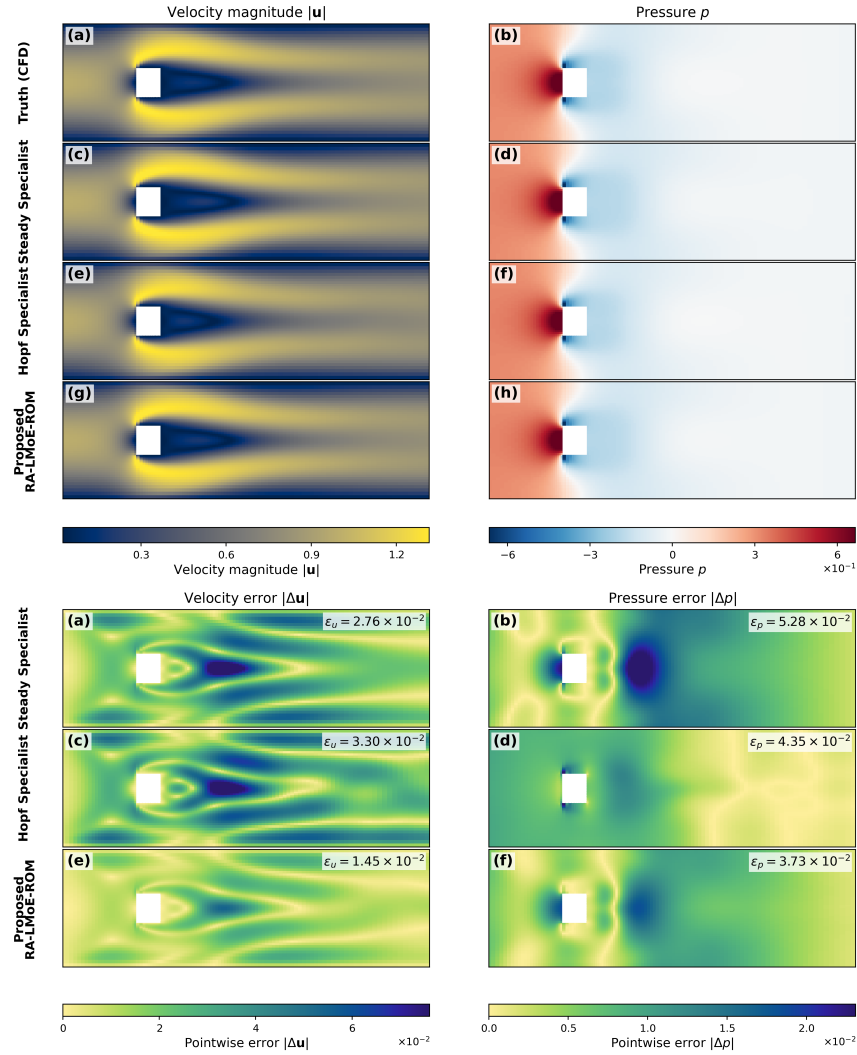


Figure 4: Representative held-out prediction in the centered-square $S-H$ overlap at $Re = 95.1$. The upper panel compares the CFD reference, Steady specialist, Hopf specialist, and T2-C prediction for velocity magnitude and gauge-corrected pressure. The lower panel gives the corresponding pointwise absolute errors.

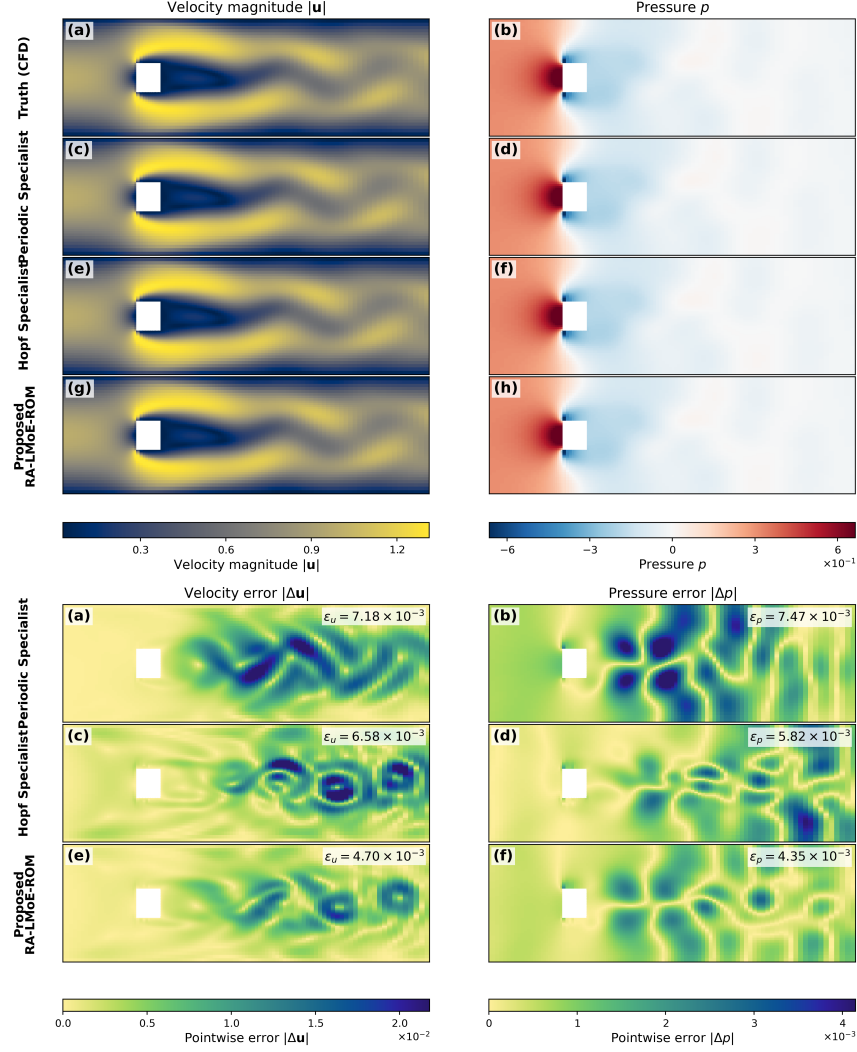


Figure 5: Representative held-out prediction in the centered-square H - P overlap at $Re = 100.5$. The upper panel compares the CFD reference, Periodic specialist, Hopf specialist, and T2-C prediction for velocity magnitude and gauge-corrected pressure. The lower panel gives the corresponding pointwise absolute errors.

H FLUIDIC-PINBALL RESULTS

The fluidic-pinball dataset contains 131 trajectories over $Re \in [10, 60]$, with 94/19/18 train/validation/test trajectories overall. A POD-DeepONet one-step baseline gives $E_u = 1.5921\%$ and $E_p = 23.3447\%$ on 4,228 held-out input-output pairs, but only 1 of 28 held-out trajectories remains finite under recursive rollout. This contrast motivates evaluation under autonomous use rather than one-step accuracy alone. The one-step baseline uses a different cadence from the matched Periodic specialist evaluation and is not a cadence-matched long-horizon accuracy baseline.

Periodic-core evaluation protocol. The $K = 32$ Periodic row in Table 2 reports the standard periodic evaluation. Periodic core is the Periodic regime’s core final-test evaluation at $K = 56$, using stride-four matched periodic chains. Snapshots are stored every 0.25 time units; taking every fourth snapshot gives an effective sampling interval $\Delta t = 1$. This aligns the evaluation with the Periodic specialist’s native history cadence and provides a longer-horizon autonomous comparison within the same regime.

Table 11: Fluidic-pinball overlap joint errors (%). Primary denotes the Steady specialist for $S-H$ and the Periodic specialist for $H-P$; the adjacent specialist is Hopf in both cases. The oracle selects its convex weight a posteriori for each evaluation window and is unavailable during prediction.

Overlap	K	Primary	Adjacent	E2 prob.	T2-C	Oracle
$S-H$	24	0.6014	3.1925	1.7511	0.5897	0.5889
$S-H$	56	0.6927	3.1193	1.7714	0.6767	0.6763
$H-P$	24	0.0990	0.5146	0.2972	0.0945	0.0928
$H-P$	56	0.1028	0.5346	0.3085	0.0978	0.0955

T2-C improves the primary single-specialist prediction in all four overlap evaluations and remains close to the window-wise convex oracle. The result is consistent across the short and long rollout horizons shown here.

I ADDITIONAL INTERNAL-ROUTING DIAGNOSTICS

Evaluation and statistics. We inspect the fixed circular-cylinder Hopf and Periodic specialists on their held-out autonomous rollout windows. The Periodic analysis uses 13 fixed windows at each of four held-out Reynolds numbers (52 windows, $K = 48$); the Hopf analysis uses five fixed windows at each of three held-out Reynolds numbers (15 windows, $K = 56$). Each statistic uses one routing record per macro-step, taken at its first integration stage, giving 2496 Periodic and 840 Hopf decisions per channel. These are macro-step statistics, not a count of all right-hand-side evaluations. Overlapping windows are not treated as independent trajectories, and no diagnostic is used for model selection.

For N decisions, let $I_{n,e}$ indicate whether routed expert e is selected, with group identity included in e . We report normalized activation frequency $f_e = \sum_n I_{n,e} / (2N)$ and mean routing mass $m_e = N^{-1} \sum_n \omega_{n,e} I_{n,e}$, retaining zero-use experts. The utilization entropy is $H = -\sum_{e: f_e > 0} f_e \log f_e / \log N_E$, where N_E counts all routed experts in the corresponding channel. Distinct pairs include group identity but ignore the order of the two selected experts. The shared/routed mixture coefficient is fixed by architecture; a shared expert is always active in the selected group, while the routed component is allocated conditionally. For both specialists here, the shared coefficient is $4/7$ and $\sum_e m_e = 3/7$. Routing mass is a mixing coefficient, not the norm of an expert’s output or its causal contribution to prediction. Figure identifiers such as $g0 : e0$ enumerate routed experts only; they are distinct from the shared-expert index 0 in Eq. (35).

Hopf routing profile. The Hopf specialist exhibits a different sparse-routing pattern from the Periodic model. It contains one group by construction and selects a fixed channel-specific pair over the evaluated windows: (e_1, e_3) for velocity and (e_2, e_4) for pressure. The same selected pairs occur at all four integration stages. Velocity expert e_3 accounts for approximately 99.93% of pooled routed mass; pressure expert e_2 accounts for 84.16%. The pressure pair nevertheless retains variable internal weighting: the mean mass assigned to the secondary pressure expert e_4 increases markedly

Table 12: Post-hoc internal-routing statistics on held-out autonomous rollout windows. Entropy is a descriptive utilization statistic, not a performance score. A fixed Top-2 pair has nonzero entropy; the two specialists also have different numbers of available groups.

Specialist	Channel	Used experts	Distinct pairs	Dominant pair (%)	H
Hopf	Velocity	2/6	1	100.00	0.387
	Pressure	2/6	1	100.00	0.387
Periodic	Velocity	18/18	33	16.27	0.929
	Pressure	14/18	16	31.25	0.784

across the three held-out Reynolds numbers. Thus, routing profiles are chart dependent: Periodic uses a broad set of configurations, whereas Hopf uses stable sparse support with nonuniform mixing weights. The sole Hopf group is a structural choice, not evidence of competition collapsing among multiple groups.

Periodic rollout-position diagnostic. Figure 6 reports routed-expert mass as a function of normalized rollout position, pooling the same 52 fixed held-out windows. This is a temporal routing diagnostic, not an estimate of physical oscillation phase. Together with Figure 3, it describes conditional allocation within the Periodic specialist without assigning physical semantics to expert identities.

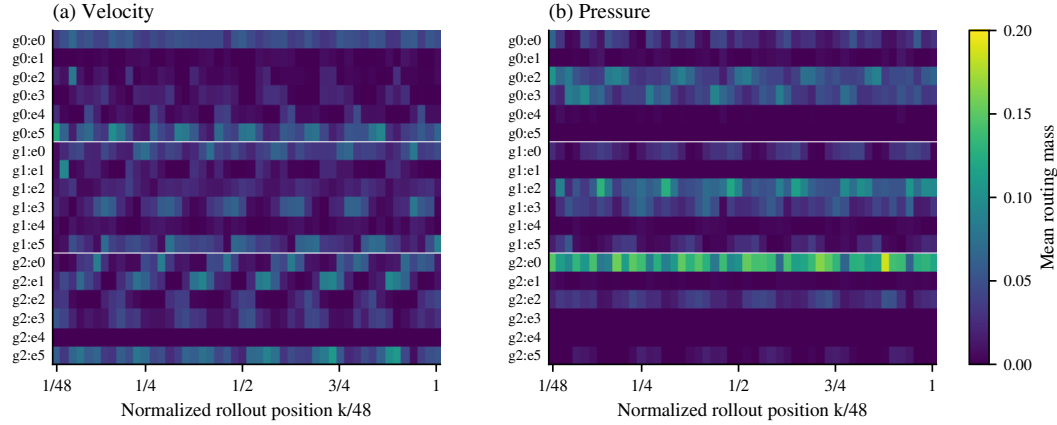


Figure 6: Routing mass across autonomous rollout position for the circular-cylinder Periodic specialist. Each column averages the corresponding macro-step decision over the fixed held-out windows; all routed-expert rows are retained. The horizontal coordinate denotes normalized rollout position $k/48$ and is not interpreted as physical oscillation phase.

J TRAINABLE AND ACTIVE PARAMETER COUNTS

Definition and verification. Table 13 reports architecture statistics for the circular-cylinder Hopf and Periodic specialists analyzed in Appendix I. Total parameters count unique parameter objects with `requires_grad=True` under the training configuration. Active parameters count the unique trainable objects actually consumed in one batch-size-one Level-1 forward, not the union across a batch, a trajectory, or all four RK4 stages. We instantiate the original model classes using recorded checkpoint configurations, verify every state-tensor shape and dtype against the archived checkpoint inventory, and trace the executed forward modules. Source hashes match those recorded in the original routing audit. This is an architecture/dispatch audit, not a replay of the pretrained checkpoint weights.

Table 13: Exact trainable and per-decision active parameter counts. Native stage 1 uses the forward options of the existing rollout evaluator; prediction only disables the optional expert-diversity diagnostic stack. Fractions use the corresponding specialist’s total trainable count.

Specialist	Total	Native stage 1	Prediction only	Native / prediction (%)
Hopf	31,810,392	22,953,160	14,165,816	72.16 / 44.53
Periodic	48,617,547	8,308,959	8,308,959	17.09 / 17.09

What is counted. Active counts include the encoder, refinement blocks, all executed routers, the selected group’s shared experts, the velocity and pressure Top-2 experts, and the learned pressure-correction head. Parameter identity deduplicates shared objects and repeated calls. The implementation evaluates every group’s channel routers even when that group’s experts are inactive; these routers are therefore included. Fixed POD bases, projected operators, scalars, and buffers are excluded. The Hopf model’s one-group router is frozen by its training definition, so its 193,755 non-trainable parameters are excluded. The separate training-loss-only normal-form head has 10 trainable parameters; it is outside the deployed specialist and the table totals.

Prediction versus diagnostic execution. The native Hopf first-stage call additionally evaluates all six routed velocity experts to form a diversity diagnostic stack. Its pressure channel remains Top-2. The Periodic prediction evaluator disables this stack. Thus native executed parameter counts need not equal the support of the sparse mixture. Turning off the stack in synthetic forward checks leaves the prediction and pressure-gate outputs bitwise unchanged; no evaluator or checkpoint was modified. All possible group/velocity-pair/pressure-pair supports have the same prediction-only count within each architecture (225 Hopf and 675 Periodic supports); dispatch checks execute every pair in both channels and every group. These are parameter counts, not measured FLOPs, runtime, or speedups.

Complete model-specific checkpoint/configuration records for the Vanilla-FNN-MoE, DataOnly-MoE, and Global MoE ablations were not available in this audit. The separate Steady diagnostic checkpoint is not established as the paper’s Steady model. We do not estimate their counts or claim parameter-matched comparisons.